BUILD - TEST - EXPLAIN

ESP32 IoT Lab Workbook

Read a DHT11 sensor and publish a simple local web response

LEVEL	FORMAT	EVIDENCE
Beginner	**2 guided labs**	**Code, wiring, checks and quiz**

This workbook supports the complete REES52 Academy course. Watch each linked lesson in the Academy, build with power disconnected, then record your evidence here.

Learner name: ______________________________ Date: _____________________

Before you build

Materials checklist

ITEM	QTY	CHECK
ESP32 development board	1	[ ]
DHT11 module	1	[ ]
10k ohm resistor	1	[ ]
Breadboard	1	[ ]
Jumper wires	5	[ ]
Data-capable USB cable	1	[ ]

Safety and setup

- Use 3.3 V logic for the DHT11 data connection.

- Never publish your Wi-Fi password in screenshots or repositories.

- Use only a trusted classroom or home network for this local server lab.

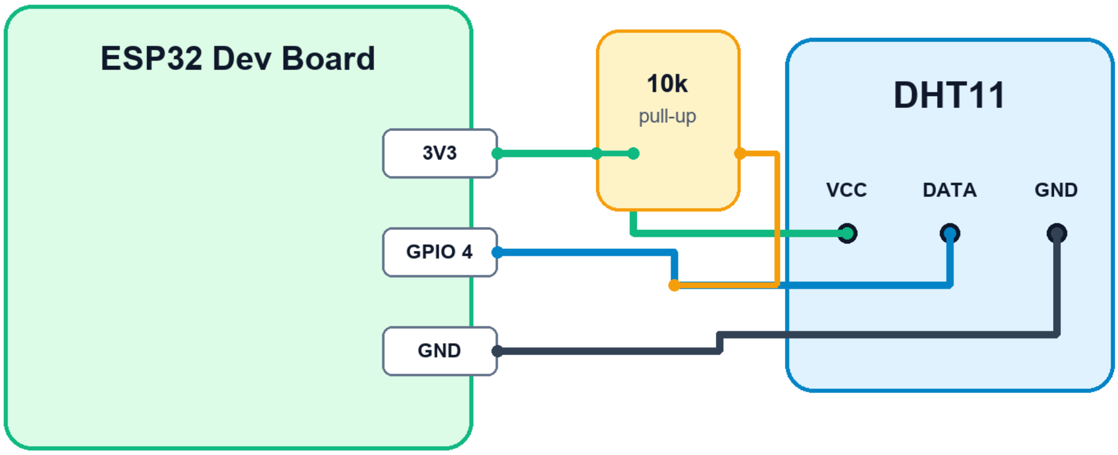
Evidence standard

A completed lab includes: a clear wiring photo, code that compiles, one observed result, one debugging note, and a short explanation in your own words.

Wiring reference

ESP32 + DHT11 Wiring

Temperature and humidity sensor lab



Use 3.3 V logic. Do not connect the DHT11 data pin to 5 V.

Verify every label against your board before applying power. Board variants can use different printed pin names.

Pre-power check

[ ] Supply voltage checked   [ ] Common ground connected   [ ] No exposed wire strands   [ ] Polarity confirmed

Lab 1: Read Temperature and Humidity

Wire a DHT11 to 3.3 V logic, print measured values and identify an invalid sensor reading.

Build sequence

1. Connect VCC to 3V3, GND to GND and DATA to GPIO 4.

2. Install the DHT sensor library used by the sketch.

3. Upload the sketch and open Serial Monitor at 115200 baud.

4. Record three readings at least two seconds apart.

Starter code

```
#include <DHT.h>

#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

void setup() {
  Serial.begin(115200);
  dht.begin();
}

void loop() {
  float t = dht.readTemperature();
  float h = dht.readHumidity();
  Serial.printf("T: %.1f C  H: %.1f %%\n", t, h);
  delay(2000);
}
```

Observation

What happened?

___

What did you change?

___

Explain why it worked:

___

Lab 2: Start a Local Web Server

Connect the ESP32 to a trusted Wi-Fi network and return a short response to a browser on the same network.

Build sequence

1. Replace the Wi-Fi placeholders only in your local copy.
2. Upload the sketch and copy the printed local IP address.
3. Open that address from a device on the same network.
4. Remove credentials before sharing screenshots or code.

Starter code

```
#include <WiFi.h>
WiFiServer server(80);

void setup() {
  Serial.begin(115200);
  WiFi.begin("YOUR_WIFI", "YOUR_PASSWORD");
  while (WiFi.status() != WL_CONNECTED) delay(500);
  server.begin();
  Serial.println(WiFi.localIP());
}

void loop() {
  WiFiClient client = server.available();
  if (!client) return;
  client.println("HTTP/1.1 200 OK\r\nContent-Type: text/plain\r\n");
  client.println("REES52 ESP32 lab is online");
  client.stop();
}
```

Observation

What happened?

____

What did you change?

____

Explain why it worked:

____

Troubleshooting map

SYMPTOM	CHECK FIRST	NEXT TEST
No output	Power, board and selected port	Run the smallest known-good sketch
Unstable readings	Loose ground or long signal wires	Print raw values and average readings
Wrong direction	Motor or output mapping	Test one output channel at a time
Upload fails	USB cable and selected board	Close Serial Monitor and reconnect

Debugging record

Issue found: ____
____

Test performed: ____
____

Final fix: ____
____

Course check

Choose one answer for each question. Submit the online quiz in REES52 Academy for scoring and feedback.

1. Which voltage is used in this wiring?

A) 3.3 V B) 12 V C) 24 V D) Mains voltage

2. Why wait between DHT11 readings?

A) The sensor updates slowly B) Wi-Fi needs charging C) GPIO 4 is analog only D) The USB cable cools down

3. What does Serial Monitor show first in the server lab?

A) The local IP address B) A certificate C) A motor speed D) A PDF

4. Where should Wi-Fi credentials be stored?

A) Only in a private local setup B) In public screenshots C) In the course title D) On the PCB silkscreen

5. Which devices can open the local server?

A) Devices on the same network B) Any device without a network C) Only the DHT11 D) Only a printer

Reflection

The concept I can now explain clearly is:

____

My next improvement will be:

____

Completion evidence

Use this final page before marking the course complete.

EVIDENCE	READY
All lesson videos watched	[ ]
Wiring built and photographed	[ ]
Starter code modified and saved	[ ]
Measured result recorded	[ ]
Online quiz passed	[ ]
Project explanation written in learner's own words	[ ]

Video source

The course lessons use tutorials published by REES52 at https://rees52.com/pages/tutorials. Links are also available inside each Academy lesson.

Course completion certificates show learning progress inside REES52 Academy. They are not a degree, licence, or government-accredited qualification.